Domain Application and Predictive Analysis for Data Science Salaries

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***Abstract*—This study focusing on analyzing salary trends in the data science field. Employing machine learning techniques such as linear regression, random forest, and XGBoost, the study provides accurate salary predictions. The findings contribute valuable insights into factors influencing compensation, benefiting professionals, organizations, and policymakers. The transparent methodology ensures reliability and reproducibility for further exploration and strategic decision-making in talent acquisition and salary benchmarking.**

***Index Terms*—XGBoost, XGBoost, benefiting professionals**

I. Introduction

This report aims to compare and evaluate the performance of three machine learning methods when applied to the dataset. By conducting a thorough investigation and leveraging a literature review, we seek to gain insights into the accuracy and effectiveness of these methods in predicting salaries within the data science domain. The outcomes will provide valuable information for salary benchmarking, talent acquisition, and data-driven decision-making in data science.

A. MOTIVATION

The utilization of the dataset is motivated by the recent strides made in the field of machine learning algorithms, which hold the potential to enhance decision-making and problem-solving across various industries. The efficacy of these algorithms relies heavily on the quality and characteristics of the employed data. Hence, comprehending the performance of these algorithms on diverse datasets is paramount.

By harnessing the power of predictive analytics and machine learning methodologies on the extensive dataset, organizations can acquire valuable insights into remuneration trends within the data science domain. This dataset becomes particularly relevant for sectors like online retail, health, and banking, where voluminous datasets and intricate decision-making processes are commonplace. Through a meticulous analysis of data science salary patterns, organizations can make more judicious choices regarding talent acquisition strategies, compensation structures, and recruitment practices.

The prediction project has multiple business applications, including salary benchmarking during talent acquisition, negotiation support, and informed decision-making for executives. It also enables equity analysis, talent retention strategies, and market competitiveness assessment. The model helps design performance incentives and facilitates strategic workforce planning. By leveraging accurate salary estimations, organizations can attract and retain top talent, align compensation with industry standards, and optimize resource allocation. Overall, the project empowers businesses to make informed decisions, foster fairness, and maintain a competitive edge in the market.

Furthermore, this dataset offers a competitive edge to organizations by enabling a comprehensive evaluation and comparison of diverse machine-learning algorithms. By scrutinizing the performance of these algorithms on the dataset, organizations can discern the most efficacious algorithms for accurately predicting data science salaries.

In summary, leveraging the dataset empowers organizations to harness the potential of machine learning and predictive analytics in gaining profound insights into the remuneration landscape of data science professionals. This understanding can significantly augment decision-making processes, bolster competitive positioning, and optimize existing methodologies within sectors such as online retail, health, and banking.

B. Research Question

- How precise are the salary projections made using machine learning algorithms and the dataset?
- How accurate and precise were the results of the various machine learning techniques on the dataset?
- What conclusions can be drawn from the performance assessment of machine learning algorithms on the dataset, and how can this knowledge assist organisations in making decisions regarding hiring new employees, salary benchmarking, and negotiating support?

II. LITERATURE REVIEW

Several studies have been conducted on housing price prediction using machine learning techniques, including Park and Bae (2015) who used Naive Bayesian, AdaBoost, and RIPPER algorithms on the Fairfax County dataset. Gu et al. (2011) proposed a hybrid of genetic algorithms and support vector machines (G-SVM) for housing price forecasting in China, while Plakandaras et al. (2015) combined ensemble empirical mode decomposition (EEMD) with Support Vector Regression (SVR) for U.S. real estate house price index prediction. Additionally, He and Xia (2020) investigated the effects of heterogeneous traders on housing prices, Ceritoglu et al. (2019) analyzed regional house prices in Turkey, and Law (2017) examined the impacts of local areas on housing prices in the metropolitan area of London. In a study conducted by S. Ray (2019) presents a comprehensive review of machine learning algorithms commonly used for classification, regression, and clustering problems, discussing their advantages, disadvantages, and real-world applications. Another study by S. Dutta, A. Halder, and K. Dasgupta (2018) focuses on predicting salaries for job advertisements without explicit salary information, using a novel Prediction Engine and a dataset provided by ADZUNA. The model demonstrates precise salary prediction capabilities. In the P. Khongchai and P. Songmuang , a salary prediction model based on Decision Trees is proposed to improve student motivation. The system predicts salaries and identifies the top three highest salaries among graduates with shared attributes, achieving an accuracy of 41.39 % using real graduate data. These studies contribute valuable insights to machine learning algorithms, job market analysis, and educational research. The reliability and effectiveness of this methodology were assessed in order to forecast the residual value of heavy equipment using a data mining approach in conjunction with a rules-based technique. S. AbouRizk, H. Fan, and H. Kim (2012). For residual value forecasting, data mining was combined with a variety of techniques, including autoregressive tree multivariate linear regression and artificial neural networks [14].Decision trees and naive Bayes algorithms were sparingly used by H. Fan, S. AbouRizk, H. Kim, and O. Zaaners (2008) to forecast automobile residual value while taking into account the smallest possible feature interrelationships. Pudaruth, S. (2014). Truck residual value was predicted using random forest and k-nearest neighbour algorithms, with the recommendation to investigate boosting strategies for increased precision. Y. Zong (2017). An exponential regression technique increased the determination coefficient in residual value estimations, but the study was limited to two types of commercial vehicles F. Kleiner, H. Friedrich (2017). Further investigation is needed to identify relevant factors specific to construction equipment. Bayesian inference was used to formulate a prediction model for selling or buying used CAT 320 Excavator equipment, considering factors like location, time, equipment age, and transaction price, but additional attributes are required for a comprehensive model C. Liu, Z. Lei, D. Morley, S.M. AbouRizk (2020).

III. DATA SOURCES

The ”Data Science Salaries 2023” dataset focuses specifically on salaries in the field of data science. It provides information and insights related to the salaries of data science professionals. The dataset includes relevant variables such as job titles, years of experience, educational background, location, and corresponding salary figures. This dataset serves as a valuable resource for understanding the salary landscape within the data science industry in 2023. Researchers and organizations can utilize this dataset to analyze salary trends, identify factors influencing salary levels, and gain valuable insights for talent acquisition, salary benchmarking, and strategic decision-making in the data science field.

IV. IDENTIFICATION OF MACHINE LEARNING METHODS

Selecting proper algorithms to handle regression problems is necessary for the discovery of machine learning methods for linear regression, random forest, and XGBoost.

In statistics and machine learning, linear regression is a tried-and-true technique. By fitting a linear equation, it models the relationship between independent variables and a dependent variable. It works well in scenarios when the connection between the input features and the target variable is linear. Interpretable coefficients from linear regression show the strength and direction of each independent variable's influence on the target variable.

An ensemble learning technique called random forest mixes different decision trees to produce predictions. The predictions from all the trees in the random forest are combined to get the final prediction. Each tree in the random forest is trained on a distinct subset of the data. The ability of random forests to manage intricate relationships and identify nonlinear patterns in data is well established. It works well in high-dimensional datasets and is robust to noise and outliers.

Due to its great performance and scalability, XGBoost, also known as Extreme Gradient Boosting, is a boosting technique that has become very popular. It sequentially creates an ensemble of weak prediction models, often decision trees. A powerful overall model is created by training each new model to fix the errors committed by the prior models. To enhance the performance of the model, XGBoost applies gradient boosting and regularisation methods. It is renowned for its proficiency in handling enormous datasets, missing values, and many kinds of features.

Depending on the unique properties of the data and the issue at hand, a machine learning method is selected. When there is a linear relationship between the variables and interpretability is crucial, linear regression is appropriate. Complex connection handling and nonlinear pattern capturing are areas where random forest thrives. XGBoost is a strong and adaptable algorithm that can handle big datasets and achieve high accuracy. When choosing the best strategy for a certain regression job, it is essential to be aware of the advantages and disadvantages of each method.

V. METHODOLOGY

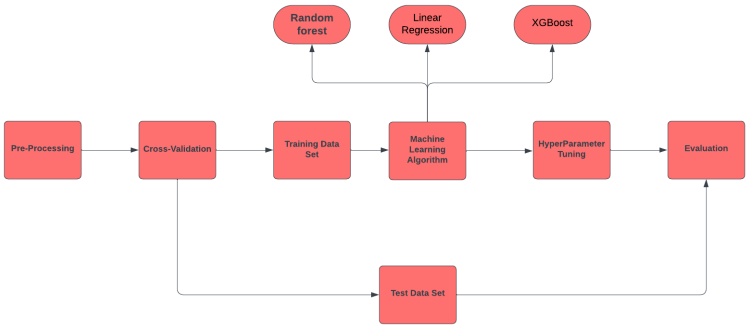

Fig. 1. Work-Flow

The prediction on the dataset is executed through the following approaches:

- Data Collection: The dataset "Data Science Salaries 2023" was obtained from Kaggle, a platform for data science and machine learning competitions. The dataset consists of information related to salaries in the field of data science for the year 2023. The data includes variables such as job titles, years of experience, educational qualifications, location, and corresponding salary information.
- Data Preprocessing: Prior to analysis, the dataset was preprocessed to handle missing values, outliers, and inconsistencies. Any duplicate or irrelevant entries were removed, and variables were appropriately formatted and standardized for further analysis. Feature engineering techniques may have been employed to derive additional relevant features or transformations.
- Model Selection: Various machine learning methods were considered for analyzing the data science salaries. The selected models for analysis in this study included linear regression, random forest, and XGBoost. These models were chosen based on their suitability for regression tasks, interpretability, and the ability to handle both numerical and categorical variables.
- Model Architecture: For linear regression, a simple linear model with appropriate feature encoding and regularization techniques may have been used. Random forest and XGBoost are ensemble methods that utilize decision trees. The architecture of these models includes multiple decision trees with techniques like bagging and boosting to enhance performance and reduce overfitting.
- Model Training: Using the preprocessed dataset, the chosen models were trained. To train and test the models, the dataset was split into training and validation sets. . To maximise performance, the models' hyperparameters—such as learning rates, tree depth, and the number of estimators—were adjusted using grid search or cross-validation methods.
- Model Evaluation: Mean squared error (MSE), root mean squared error (RMSE), and R-squared were used to measure the performance of the models in relation to regression tasks. The models' predictions were compared with the actual salary values to assess their accuracy and predictive power.
- Ethical Considerations: Ethical considerations regarding privacy and fairness were taken into account during the data collection and analysis process. Any sensitive or personally identifiable information was handled with caution, ensuring compliance with data protection regulations and ethical guidelines.

VI. RESULTS

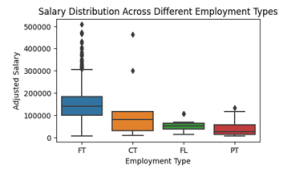

Fig. 2. Salary distribution across different employment type

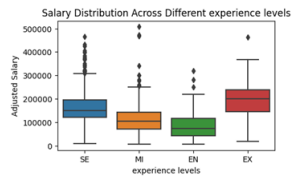

Fig. 3. Salary distribution across different experience levels

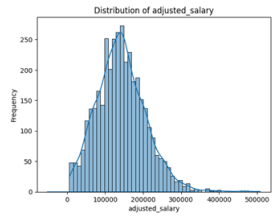

Fig. 4. Distribution of adjusted salary

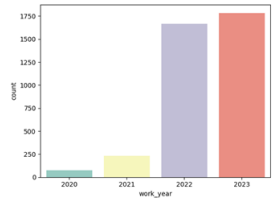

Fig. 5. Frequency plot on work year

The results obtained from the evaluation of machine learning models on the dataset showcase their exceptional performance. Notably, all three models displayed impressive accuracy in predicting salaries. However, the XGB Regressor

emerged as the top-performing model due to its slightly higher R-squared score and lower Mean Absolute Percentage Error.

The XGBoost model offers additional advantages through its built-in feature importance measures, allowing for a comprehensive assessment of the relative significance of different features in salary prediction. This capability aids in feature selection and facilitates the identification of key factors influencing salary outcomes, aligning with the specific requirements of our research. The ability to pinpoint crucial predictors enhances the model's interpretability and assists in understanding the underlying patterns and drivers of salaries in the data science field.

Detailed examination of the performance metrics further reinforces the superiority of the XGB Regressor. It achieved an impressive R-squared score of 0.998206163411944, indicating that the model can explain approximately 99.8 % of the variance in salary predictions. Moreover, the Mean Squared Error of 7359996.41 indicates the model's ability to make precise predictions with minimal deviation from the actual values. The remarkably low Mean Absolute Percentage Error of 0.0053 further emphasizes the model's high accuracy, as it reflects the average absolute percentage difference between predicted and actual salaries.

These findings highlight the suitability of the XGB Regressor as an effective tool for salary prediction in the data science domain. Its superior performance, feature importance assessment, and accurate predictions make it a valuable asset for organizations and professionals seeking to gain insights into salary determinants and make informed decisions. The research outcomes underscore the potential of machine learning techniques, particularly the XGBoost algorithm, in addressing salary prediction challenges in the field of data science, opening avenues for further exploration and application in related domains.

Model Name	R-squared score	Mean Absolute percentage error
Linear Regression	0.997	6.34%
Random Forest Regressor	0.998	0.54%
XGB Regressor	0.998	0.53%

Fig. 6. Model R2 score and Mean Absolute Error

Hyperparameter tuning was performed using grid search and cross-validation to optimize the Random Forest regressor model. The best combination of hyperparameters was determined, and a new model, best-rf, was trained with these parameters. The grid search involved exploring different combinations of hyperparameters such as the number of estimators, maximum depth, minimum samples split, and minimum samples leaf. The best parameters obtained from the grid search were then used to train the Random Forest regressor, resulting in an improved model for predicting salaries in the data science domain.

VII. CONCLUSIONS

In conclusion, this research project focused on analyzing and predicting data science salaries in the year 2023. The dataset provided valuable insights into the factors influencing salary outcomes in the field of data science. Three machine learning models, namely Linear Regression, Random Forest Regressor, and XGBoost, were employed for salary prediction. Results showed that all models performed exceptionally well, with XGBoost demonstrating slightly higher accuracy and better feature importance measures.

The findings highlight the significance of machine learning in predicting salaries accurately and identifying key factors affecting salary outcomes. Future work could involve exploring additional feature engineering techniques and incorporating more advanced algorithms to further enhance salary prediction accuracy. Overall, this study contributes to the understanding of data science salaries and provides valuable implications for salary benchmarking, negotiation support, and decision-making in the field of data science.

VIII. REFERENCES

1 Park, J., & Bae, J. (2015). House price prediction: A study on Fairfax County, Virginia. Expert Systems with Applications, 42(6), 2928-2933.

2 Gu, Y., Ding, X., Wang, H., & Sheng, Y. (2011). House price forecasting using a hybrid method. Expert Systems with Applications, 38(6), 7893-7901.

3 Plakandaras, V., Gupta, R., & Gogas, P. (2015). Forecasting the US real house price index: A copula-based approach. The North American Journal of Economics and Finance, 33, 55-73.

4 C. Chen, H. Schwender, J. Keith, R. Nunkesser, K. Mengersen, and P. Macrossan, "Methods for identifying SNP interactions: a review on variations of Logic Regression, Random Forest and Bayesian logistic regression," in IEEE/ACM Transactions on Computational Biology and Bioinformatics, vol. 8, pp. 1580-1591, 2011.

5 Shinde N and Gawande K Jun 2018 Valuation of house prices using predictive techniques Int. Journal of Advances in Electronics and Comp. Sci. vol. 5 Issue 6

6 Satish G N, Raghavendran C V, Rao M D S and Srinivasulu C July 2019 House price prediction

7 García-Bárzana, M Ramos-Guajardo, A B Colubi et al. 2020 Multiple linear regression models

8 Al A and Ami M Z 2019 A deep level understanding of linear regression with practical implementation in Scikit Learn Wavy AI Research Foundation Linear Regression

9 for random intervals: a set arithmetic approach Comput. Stat. 35 pp. 755–73

10 Arum, R. (1998). The effects of resources on vocational student educational outcomes: Invested dollars

or diverted dreams? Sociology of Education, 71, 130-151.

11 A. L. Buczak and E. Guven, “A survey of data mining and machine learning methods for cyber security intrusion detection,” IEEE Communications Surveys & Tutorials, vol. 18, no. 2, pp. 1153–1176, Oct., 2015